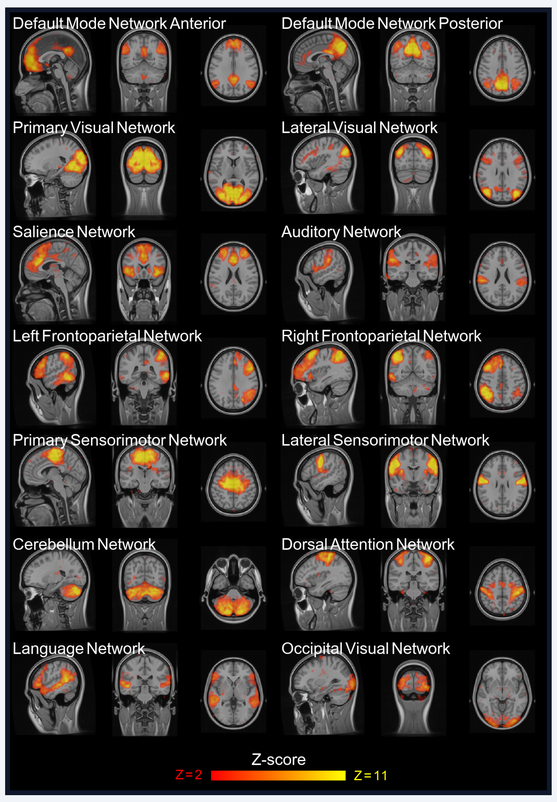
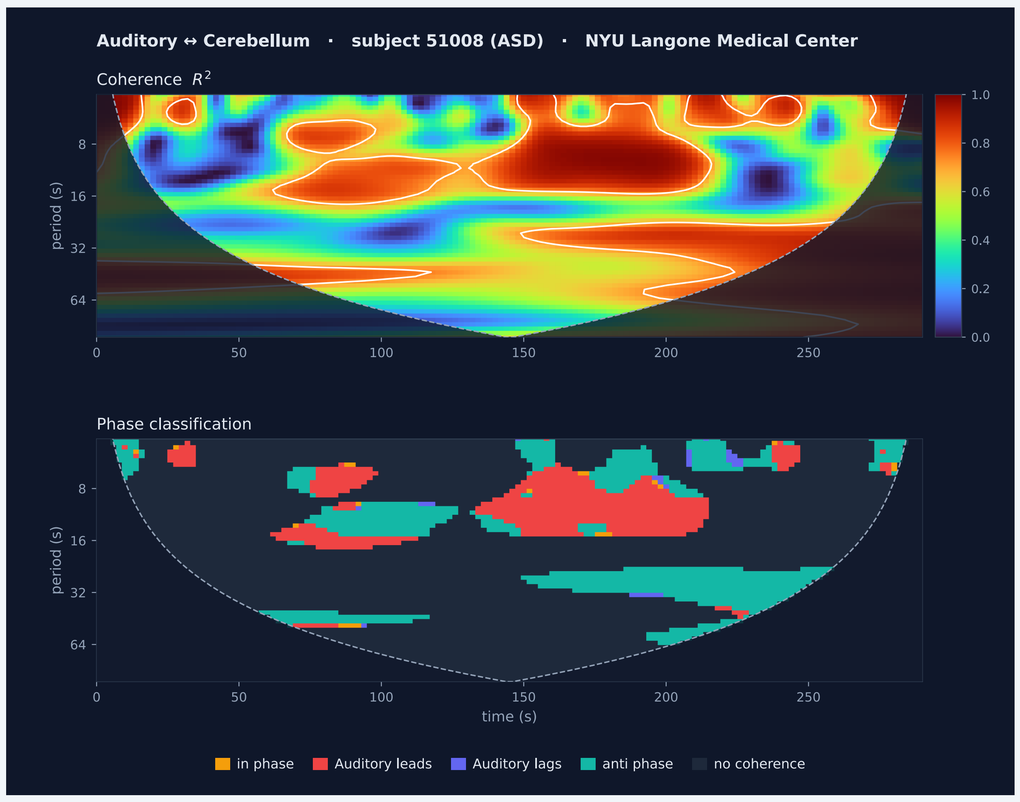
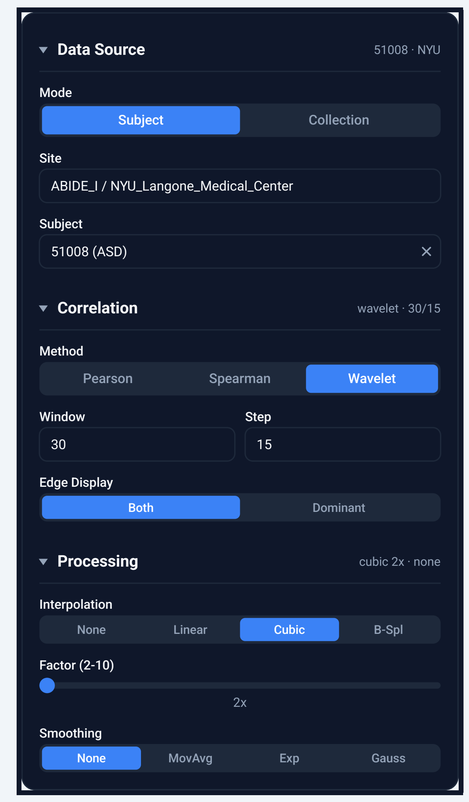
Interactive Visualization of Dynamic Functional Connectivity in the ABIDE Autism Dataset

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System overview. **Left:** The 14 resting state networks from group ICA. **Centre:** Morlet wavelet coherence for one network pair, a localised correlation coefficient in time and frequency space, alongside a phase classification map [3,8]. The correlation metric is the ratio of *leads* to other classifications within a sliding time window across all frequencies. **Right:** Dynamic visualization controls. Every change recomputes the graph in real time.

Introduction

Resting state fMRI is the most studied imaging method in autism spectrum disorder research [1].

Functional connectivity is conventionally computed over an entire recording, resulting in a **single static connectivity matrix**. However, over the course of a recording, connectivity patterns can vary as the mind wanders. **Dynamic functional connectivity** captures the information that static correlations discard [3].

Therefore, a visualization of dynamic connectivity patterns can help clinicians understand a diagnostic feature, rather than trust an opaque prediction.

Method

900
recordings
417 ASD / 483 HC

14
resting state networks
from group ICA

Recent work preprocessed **ABIDE I+II** using independent component analysis into **resting state networks** [2]. Functional connectivity between two networks is quantified by correlating their time series. Applied over a **fixed-length sliding window**, this produces a series of matrices that describe the dynamic functional connectivity [4,5].

THREE CONNECTIVITY MEASURES

Pearson **Spearman**

Wavelet lead coherence

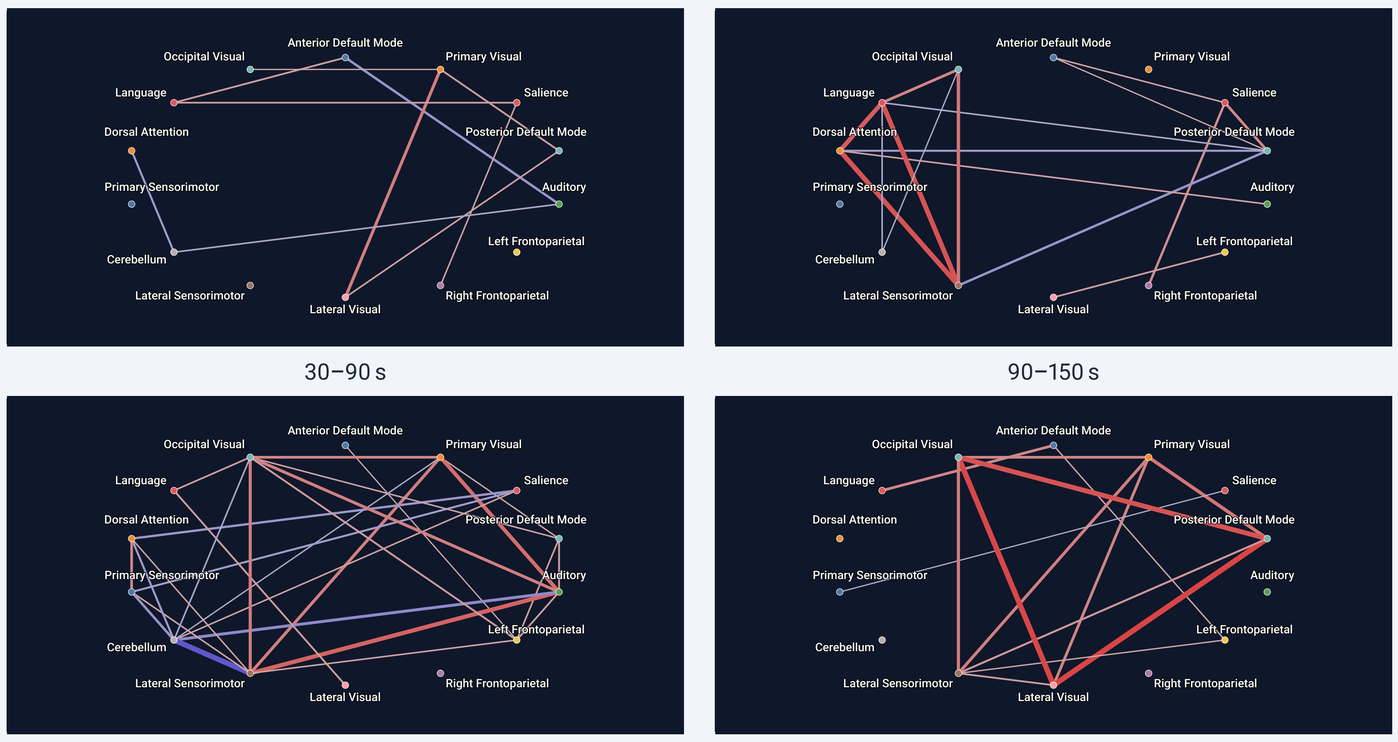
Pearson and Spearman correlation produce symmetric matrices in the range $[-1, 1]$. The third is a directional (asymmetric) measure, called **wavelet lead coherence**. It reports the fraction of time and frequency points at which one network leads the other, so the resulting matrix is asymmetric [6,7].

Conclusions

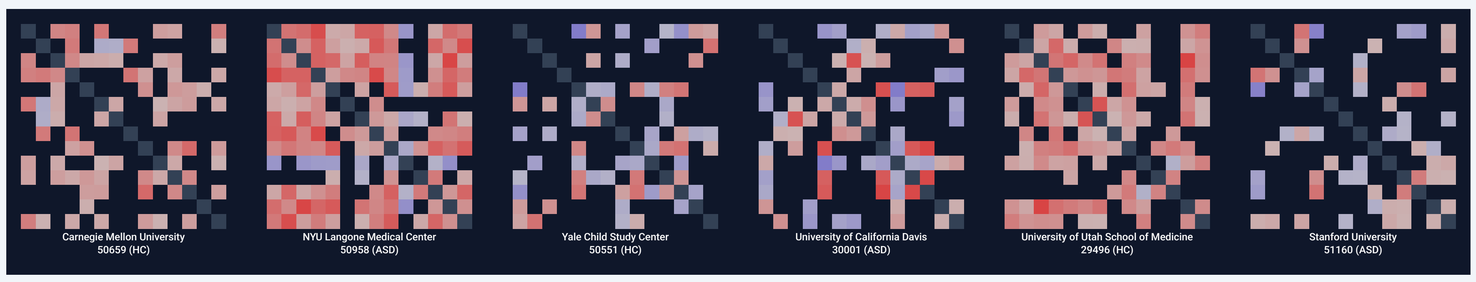
Dynamic functional connectivity is often studied as a diagnostic marker, yet its high-dimensional nature limits interpretability. Existing tools visualize dynamic connectivity statically, so the dynamic patterns are difficult to observe.

Our contribution is a novel visualization tool for observing dynamic functional connectivity patterns between RSNs, applied to the **ABIDE I+II** dataset. Planned work adds a quantitative comparison of ASD against control subjects.

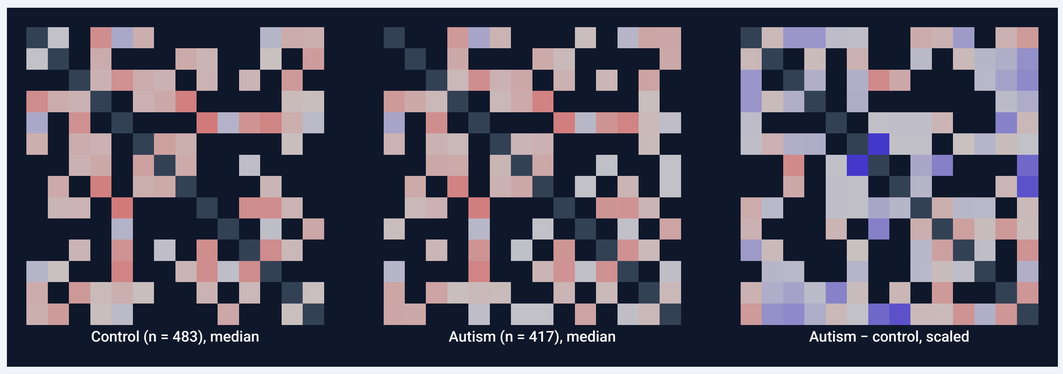
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Graph view. One recording as four successive 60 s windows. The graph view places the resting state networks on a ring and draws each frame's connectivity matrix as edges. The color of an edge encodes the correlation value on a blue to red scale and its thickness encodes the absolute magnitude. A configurable threshold hides weak edges and denoises the visualization. For asymmetric correlations the edges carry arrows showing directionality. Hovering over a node isolates its edges, and any change in the parameters recomputes the graph in real time. A smoothing algorithm can be applied to smooth sharp transitions, and interpolation adds intermediate frames for a smoother and longer visualization.



Individual subjects. A complementary overview pane reduces each subject to a single correlation matrix over the entire recording, shown as a heatmap or a small graph, labelled with its acquisition site and subject number. Identifying site variation and measurement error induced outliers is especially important, as it is a known confound in multisite rs-fMRI.



Class comparison. Class aggregate connectivity matrices and connectivity difference. Using the mean or median over all ASD and control subjects, an individual's dynamics can be compared against either class. Class aggregates differ slightly, so the difference matrix is scaled for visibility.

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